Acne Detection and Cross-Domain Classification — Report

1 Overview

This project tackles two connected problems on dermatology imagery: (1) detecting individual acne lesions in ACNE04 with bounding-box object detection, and (2) transferring an acne classifier trained on ACNE04 patches to DermNet, a clinical skin-condition dataset with a substantial domain gap. Part 1 measures localization quality on a single dataset; Part 2 measures how well learned acne features generalize across imaging conditions, framing, and label distributions.

2 Evaluation metrics

For detection: IoU $= |B_{\text{pred}} \cap B_{\text{gt}}|/|B_{\text{pred}} \cup B_{\text{gt}}|$, Precision $= TP/(TP+FP)$, Recall $= TP/(TP+FN)$; for DermNet: $F_1 = 2PR/(P+R)$. I report **mAP** (COCO-style), precision/recall, and mean matched IoU@0.5 for detection; **Accuracy**, **F1**, and **AUROC** for classification.

3 Part 1: Object Detection on ACNE04

3.1 Models

I trained two detectors spanning classic vs. modern: **Faster R-CNN** (Ren et al., 2015) — two-stage, ResNet-50 FPN, COCO-pretrained — and **YOLOv5s** (Ultralytics) — single-stage YOLO lineage. YOLOv5 was chosen over heavier detectors (e.g. DINO-DETR) for efficiency and strong community baselines on small objects. The Roboflow YOLO head follows the export's class list; Faster R-CNN uses a two-class COCO head (background, acne). For matched IoU and the greedy P/R summary, predictions are treated as one lesion class so both detectors are compared on the same boxes.

3.2 Data, training, evaluation

ACNE04 (Roboflow): YOLO + COCO exports for fair splits. Patched `data.yaml` with absolute paths. **YOLOv5:** 50 epochs, 640px, batch 16, `yolov5s.pt`. **Faster R-CNN:** 10 epochs, SGD lr 0.005, momentum 0.9, wd 5×10^{-4}, batch 4, two classes. Faster R-CNN mAP and greedy Precision/Recall/mean matched IoU@0.5 are on the **COCO test** split (`torchmetrics.MeanAveragePrecision`, conf >0.25, IoU ≥ 0.5). YOLO mAP@0.5, mAP@0.5:0.95, Precision, and Recall are the **last-epoch validation** metrics from Ultralytics `results.csv` (YOLO's val split in `data.yaml`), while YOLO mean matched IoU@0.5 is recomputed on the **same COCO test images** as Faster R-CNN via `torch.hub` for a fair localization comparison.

3.3 Visualization

Figure 1 shows six test images per detector. YOLOv5 shows the export's class names where present; Faster R-CNN shows red boxes with confidences mostly 0.6–0.95 (higher test recall in Table 1); YOLOv5 achieves slightly higher mean matched IoU@0.5 on the shared COCO test pass.

4 Part 2: Cross-Domain Classification

4.1 Patches, baseline, DermNet setup

Positive/negative 64×64 patches from ACNE04 boxes (equal counts); box stats checked for scale. **Baseline:** ResNet-18 ImageNet (He et al., 2016), 10 epochs, Adam 10^{-4}, flips + color jitter, 80/20 split. DermNet binary: `Acne and Rosacea Photos` vs. all other folders (∼1:22 imbalance). `ImageFolder` order differs from ACNE04 (`negative/positive` ⇒ acne index 1); evaluation maps label spaces explicitly.

4.2 Domain adaptation

(1) `model_cls2`**:** InceptionResnetV1 VGGFace2 (`facenet-pytorch`), 160×160, stronger aug. (45° rot., jitter), class weights $[1.0, 5.0]$. (2) `model_cls3`**:** Per-channel histogram matching: each patch is matched to a random reference image from the DermNet **train** split (acne folder vs. pooled non-acne folders), then FaceNet is retrained on the matched patches. (3) `model_cls4`**:** Pseudo-labels from `model_cls3` on **20 unlabeled DermNet train** images (10 from the acne-related folder + 10 from other folders, shuffled); keep top-K=5 confident predictions per predicted class; 5 epochs fine-tune at 10^{-5}.

4.3 Evaluation

All checkpoints evaluated with `evaluate_on_dermnet`: resize per model, P(acne), F1-optimal threshold from PR curve; Accuracy, F1 (acne positive), AUROC.

5 Results

Table 1 summarizes Part 1. mAP@0.5 is similar across detectors (∼0.21–0.23) but is *not* measured on identical splits for YOLO vs. Faster R-CNN (see caption). On the shared test images, mean matched IoU@0.5 is nearly tied. Faster R-CNN shows higher **test** recall under the greedy matcher (0.45 vs. 0.28 for YOLO's val recall), but the val/test asymmetry means that gap is only directional, not definitive.

Table 2 reports Part 2 on the full DermNet binary test set (accuracy, F1, AUROC, and F1-optimal threshold from the PR sweep in `evaluate_on_dermnet`).

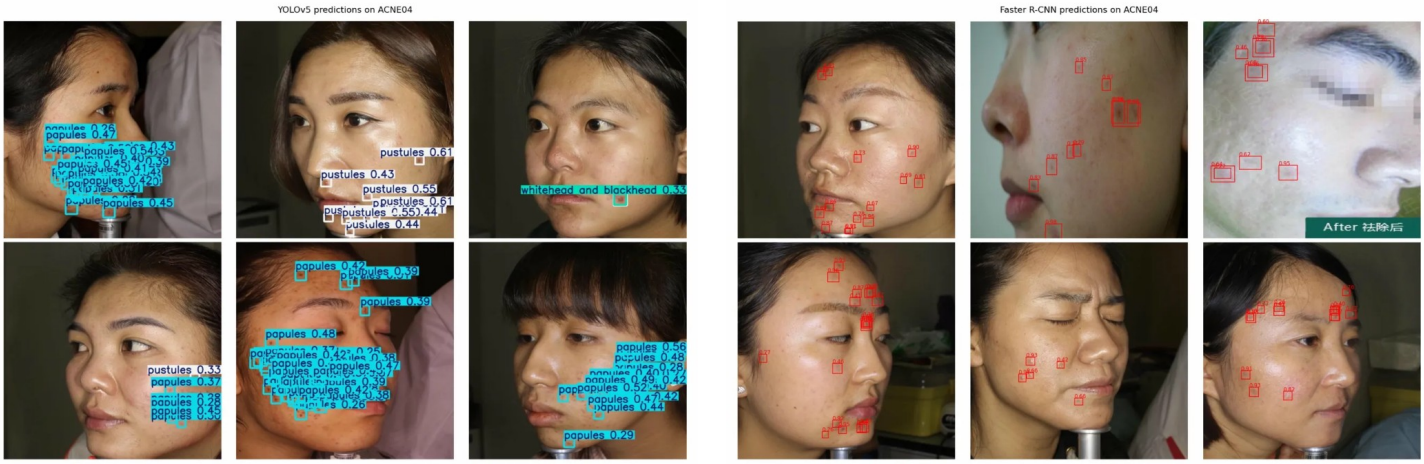
5.1 Grad-CAM

Grad-CAM on `model_cls2`–`model_cls4`: 10 DermNet test images each (5 acne / 5 non-acne), FaceNet `block8`, acne class. Figure 2: 2×5 panels; `True/Pred` with confidence; green/red = correct/incorrect; warm colors = high contribution to acne activation.

6 Analysis (reflection)

Transfer headline. AUROC stays in a narrow band (roughly 0.58–0.63); no adaptation clearly beats the ResNet-18 baseline in a way I would trust at single-seed (re-runs shifted AUROC by ∼0.04 in development). Histogram matching nudges AUROC slightly above baseline here, but the margin is small enough to be noise.

Accuracy vs. AUROC. Pseudo-labeling reaches the highest accuracy (0.67 vs. 0.53 baseline) yet AUROC remains flat; the F1-optimal threshold jumps to 0.78, so the model rarely



a YOLOv5 **b** Faster R-CNN

Figure 1: Qualitative detection on ACNE04 (six test images each).

Metric	YOLOv5	Faster R-CNN
mAP@0.5	0.2111	**0.2302**
mAP@0.5:0.95	**0.0520**	0.0507
Precision@0.5	**0.3068**	0.2851
Recall@0.5	0.2840	**0.4467**
Mean IoU@0.5 (matched)	**0.6323**	0.6218

Table 1: Detection on ACNE04. Faster R-CNN rows are on the COCO **test** split. YOLO mAP/P/R are from the training run's **val** split (`results.csv`); YOLO mean matched IoU@0.5 is on the COCO **test** split (same images as Faster R-CNN).

predicts acne unless nearly certain—most "accuracy" comes from the easy non-acne majority under 1:22 imbalance. This is threshold gaming, not a cleaner ROC.

Grad-CAM / failure modes. Across 30 panels (10 images × three FaceNet checkpoints), attention repeatedly overlaps the **DermNet.com watermark**. The shortcut persists after histogram matching and pseudo-labeling, which helps explain why adaptation did not produce a robust AUROC gain: all runs can exploit the same artifact. Low F1 (∼0.19) across conditions reflects imbalance plus the patch→full-image gap.

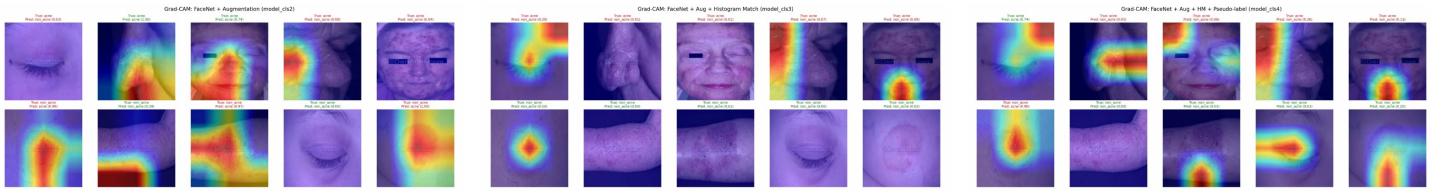
Next steps. Inpaint or remove watermarks; report PR-AUC; balanced sampling or focal loss; DANN/CORAL on features; multi-seed runs before claiming method rank order.

References

1. Ren et al. (2015). Faster R-CNN. https://arxiv.org/abs/1506.01497
2. Redmon & Farhadi (2018). YOLOv3. https://arxiv.org/abs/1804.02767
3. Lin et al. (2014). MS COCO. https://arxiv.org/abs/1405.0312
4. He et al. (2016). ResNet. https://arxiv.org/abs/1512.03385
5. Selvaraju et al. (2017). Grad-CAM. https://arxiv.org/abs/1610.02391

Model	Acc.	F1	AUROC	Thr.
Baseline (ResNet-18)	0.5313	0.1917	0.6174	0.0138
FaceNet + Aug.	0.5632	0.1756	0.5832	0.3313
Aug + Hist. match	0.6590	0.1909	0.6275	0.4257
Aug + HM + Pseudo	0.6734	0.1937	0.6246	0.7781

Table 2: DermNet binary test (acne vs. non-acne), one Colab run (single seed).



a `cls2` **b** `cls3` **c** `cls4`

Figure 2: Grad-CAM on DermNet (FaceNet `block8`, acne class).